

son who needs to look at keys when typing, TREWgrip offers visuals of keys on the top of the product with LEDs to indicate what key you pressed below. Although we still think of this as a stop-gap measure in the UI race to perfection, it certainly has the potential to improve the one experience of touch devices we all hate.

If typing on the back of your phone is too weird an idea, then there're always alternatives to on-screen keyboards. Minuum, for example, is a single line keyboard that relies heavily on auto-correct to get typing right. It's currently available on the Google Play store for android users, but it goes way beyond that. The aim of the company is to give you the ability to make almost any surface a keyboard. Using a smart ring, or a wristband, or a game controller, or even your own forearm, you can input text that's wirelessly transferred to the device you've paired with. You can read more about the proposed technology at minuum.com.

Plus, Google is also thinking along similar lines for project Glass, and is experimenting with a laser projection capability that would allow pretty much any surface to become a keyboard, or a text input device. Perhaps simple laser displays and Minuum-type keyboards will allow us to just type out stuff anywhere, with our phones never even being visible.

Such research is of the utmost importance, because we truly believe that the old Icon-based UI just has to go, but however also realise that touch or text just isn't going anywhere in a hurry, so these are technologies that will bridge the gap and allow us to experiment with UIs that are actually made for devices that are used on the go, instead of just scaling down UIs that were built for large, immovable displays. But what about said displays? Should we get rid of those altogether?

No screen

With everyone selling phones based on screen sizes, and actually trying to advertise an almost tablet-sized 6-inch screen as



Smart watches are all the rage now

bigger is better on a phone, we're sort of regressing. Some manufacturers are just trying to get us back into the old LCD monitor war days where inches were everything. At the risk of being risqué, it's not how many inches you have people, it's how you use them! This whole bigger is better race seems to be encouraged by the fact that people are giving up, and opting for phablets because they have realised that they just need a bigger screen for some tasks. For this, some of us are willing to look like dorks with large Bluetooth devices in our ears, or worse, willing to hold up a tablet-sized device to our ears to make phone calls. Yes larger devices have larger screens, and videos look good on them, as do web pages, but they look funny in your pocket, ridiculous when held to your face, have huge batteries that take longer to charge, but the same time to run out, are heavy, and are just not the answer.

Google Glass and other wearable screens are one way ahead, because you can project virtual screens (the virtual screen appears to be a 15-inch display, or 17-inch, etc) when you need them – to watch videos or look at a large web page, etc. However, these glasses also allow you to keep things limited to your peripheral vision, so you're updated with your notifications, but not at the cost of being distracted with what's in front of you. They have audio features also, and allow you to do tasks with spoken commands, or take calls without touching your phone. The problem is they look too dorky, but that's also fixable. What isn't fixable is that some idiot is eventually going to mow down school children crossing a road because he wore his glasses while driving and forgot to turn notifications off, or worse, was talking to his co-passenger and said, "Ok, you ass, you tell me what's the fastest way to MG road?" and his glass heard "Ok Glass, what's the fastest way to MG Road" and promptly popped up a map, or started talking in his ear, and distracted him long enough to kill someone!

Apart from this, voice recognition in general is OK, but only in situations where you're alone or won't be disturbing anyone, or perhaps when you don't mind people overhearing your commands.

Divergence

Another interesting area that's become popular is smart watches,



Meet Minuum, the single-line QWERTY

which basically allow you to do all of your basic tasks on your wirelessly connected watch, using your phone's processing power, and since this is a second device, it works well for all applications that don't need a large screen. If a smart watch is paired with a phablet, you can have the best of both worlds, in a way, by having two separate devices. If people start selling these as bundled products – Apple might sell an iWatch and iPhone combo, or allow the iWatch to bring phone functionality to the iPad – then this does become a way of separating UIs for varying tasks. To make or receive a phone call, you barely need anything more than the two line displays of yesteryear. To play a game or a movie however, you need the high resolution displays of today. Perhaps the future is really diverging products as per their usage scenario, and designing custom UIs that are purpose built, and thus perform those set tasks better.

Of course what we're facing today is sort of divergent – your phone or tablet for fun and your laptop for serious work – so perhaps there really is no way forward, and we've reached the pinnacle of human ingenuity in terms of mobile devices... No, we don't really believe that either! Why on earth would Google be spending so much time researching ways to do away with screens completely. It's not just to sell a product like Glass, it's a valiant attempt to finally replace the ageing icon-based system.

There are hundreds of more ideas for both divergent and convergent devices, and the UIs that will rule will be decided by what form factors of devices we are drawn to. At our end, we just hope that more powerful and more flexible devices will win in the end, so that we can finally be rid of the need to buy and operate multiple form factors just to get through an average life. 